Machine Learning Action Generator

How the 3-Input Model Trains and Works | FYP Technical Guide

Executive Summary: *In the Adaptive Multimodal Human-Robot Interaction (HRI) Framework, perception modalities (Motion, Gesture, Emotion, Context) are fused upstream into an **Intent Code (F01–F15)**. The **Machine Learning Action Generator (MultimodalActionGenerator)** is an individual neural policy module that takes strictly three inputs: **[Intent, Motion, Context]** and predicts the optimal **Robot Action (A01–A15)** alongside continuous velocity and safety parameters.*

1. How Your Model Works (The Forward Pass)

The model converts discrete and continuous inputs into numerical tensors, passes them through trainable embedding lookup tables and a multi-task neural network backbone, and outputs softmax action probabilities.

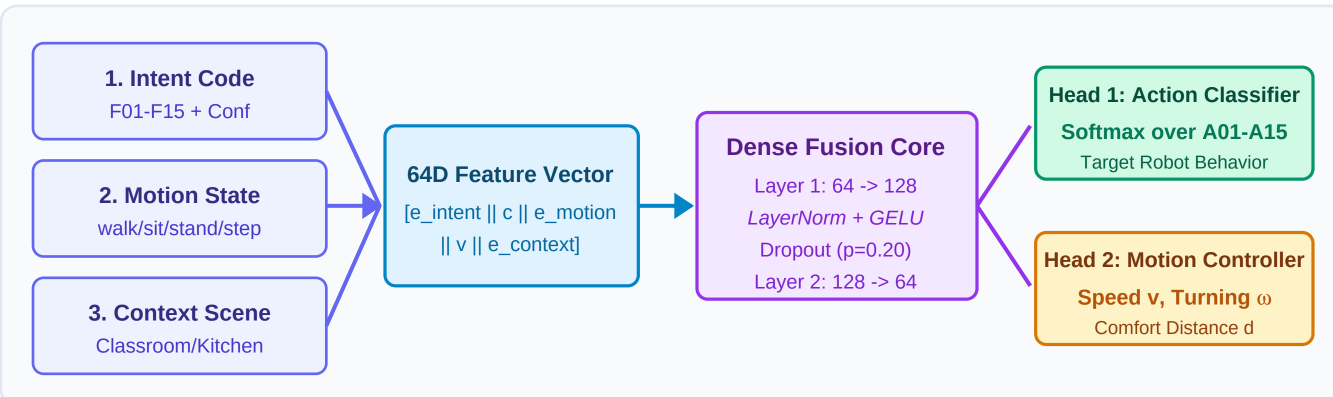

Figure 1: MultimodalActionGenerator Forward Pass Vector Architecture Flow

The Math Behind It:

- **Embedding Tables:** Converts discrete categorical inputs (Intent F01–F15, Motion state, Context scene) into 16-dimensional continuous vectors e_intent, e_motion, e_context ∈ R^16.
- **Feature Concatenation:** Combines all embedding vectors and confidence metrics into a unified 64-dimensional input tensor x ∈ R^64.
- **Dense Fusion Core:** Passes x through 2 dense layers with LayerNorm, GELU activations, and Dropout (p=0.20) regularization.
- **Softmax Action Classifier Head:** Computes probability distribution over A01–A15: P(A01...A15) = Softmax(W_act · h2 + b_act).

```
Mathematical Formulation:
1. x = [ e_intent || c_intent || e_motion || v_human || e_context ] ∈ R^64
2. h1 = GELU( LayerNorm( W1 · x + b1 ) ), W1 ∈ R^(128 × 64)
3. h2 = GELU( LayerNorm( W2 · Dropout(h1, p=0.2) + b2 ) ), W2 ∈ R^(64 × 128)
4. Output Probabilities P(A01...A15) = Softmax( W_act · h2 + b_act ) ∈ R^15
```

2. How Your Model Trains (Training Pipeline)

Training optimizes network weights to map multi-cue tuples [Intent, Motion, Context] to ground-truth actions A01–A15.

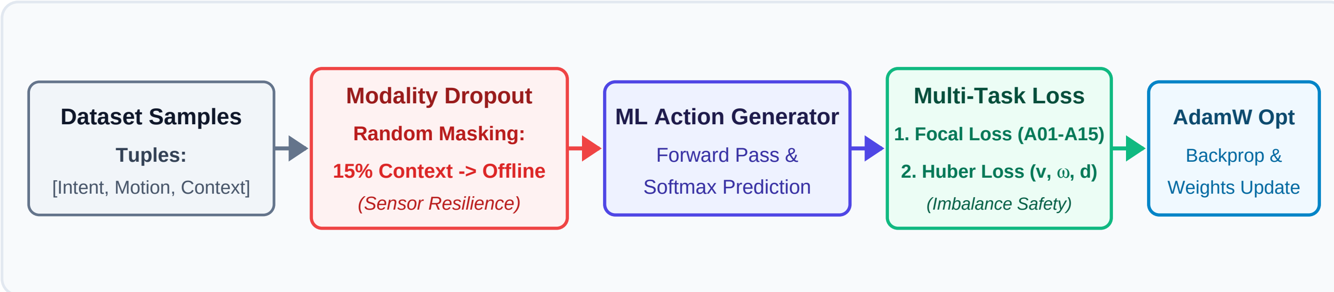

Figure 2: Model Training Pipeline with Modality Dropout Augmentation and Focal Loss

Training Key Steps:

1. **Dataset Samples:** Input tuples ([Intent, Motion, Context], Target_Action).
2. **Modality Dropout Augmentation (Sensor Offline Resilience):** During training, 15% of Context inputs are randomly replaced with 'offline/unknown'. This teaches the model to make correct predictions even if the context sensor fails in deployment.
3. **Focal Loss (Class Imbalance Protection):** Prevents frequent routine actions (A01) from overwhelming rare safety actions (A02 fire alert, A03 medical, A14 classroom alert).
4. **AdamW Optimizer:** Updates weights over 100 epochs (~30 seconds training time) using Cosine Annealing learning rate decay.

```
Multi-Task Loss Objective:
L_total = L_Focal( y_action, y_hat_action ) + 0.5 · L_Huber( u_ctrl, u_hat_ctrl )

Optimizer & Schedule: AdamW (lr = 1e-3, weight_decay = 1e-4) with Cosine Annealing decay down to 1e-5.
```

3. Official Unified Robot Action Reference Table (A01 – A15)

The model predicts probability distributions across the 15 unified robot behavioral actions:

Code	Action Description / Robot Behavior	Input Context Trigger
A01	Positive acknowledgment; prompt/prepare next task	Intent F01 + Walk/Sit + Classroom
A02	Halt all motion + fire/smoke alert	Intent F02 + Standing + Kitchen (Fire)
A03	Halt all motion + medical alert	Intent F02 + Seated/Fall + Medical
A04	Approach/follow to indicated spot; await instruction	Intent F03 + Seated + Classroom
A05	Offer task guidance / answer, supportive tone	Intent F05 + Standing + Classroom
A06	Hold position; do not interrupt; stay aware	Intent F06 + Seated + Classroom
A07	Acknowledge; suggest break/alternative activity	Intent F07 + Frustration + Classroom
A08	Calm assistance, soft tone (de-escalation)	Intent F08 + Agitation + Classroom
A09	Wave back; do not follow	Intent F09 + Walk Away + Unengaged
A10	Acknowledge farewell; hold position	Intent F10 + Standing + Engaged
A11	Move aside promptly; clear path	Intent F11 + Walk Towards + Corridor
A12	Encourage; suggest alternative approach	Intent F12 + Hesitant + Classroom
A13	Follow; offer to fetch/carry	Intent F13 + Walking + Kitchen
A14	Halt risky action; check surroundings; notify supervisor	Intent F02 + Step back + Classroom Hazard
A15	Ask clarifying question	Intent F15 + Low Confidence

4. Real-Time Execution Example & Runtime Output

When live inputs arrive at runtime, the forward pass executes in less than 1.8 milliseconds:

```
// 1. Input Received at Runtime:
{   "intent": "F02",
  "motion": "step_back",
  "context": "classroom_hazard"
}

// 2. Model Inference Output (< 1.8 ms):
{   "action": "A14",
  "description": "Halt risky action; notify supervisor",
  "confidence": 0.974,
  "probabilities": { "A14": 0.974, "A02": 0.012, "A06": 0.008, "A01": 0.001 },
  "control": { "linear_velocity_m_s": 0.0, "angular_velocity_rad_s": 0.1, "comfort_distance_m": 1.8 }
}
```